Class: Algorithms & Data Structures Semester: Spring 2025

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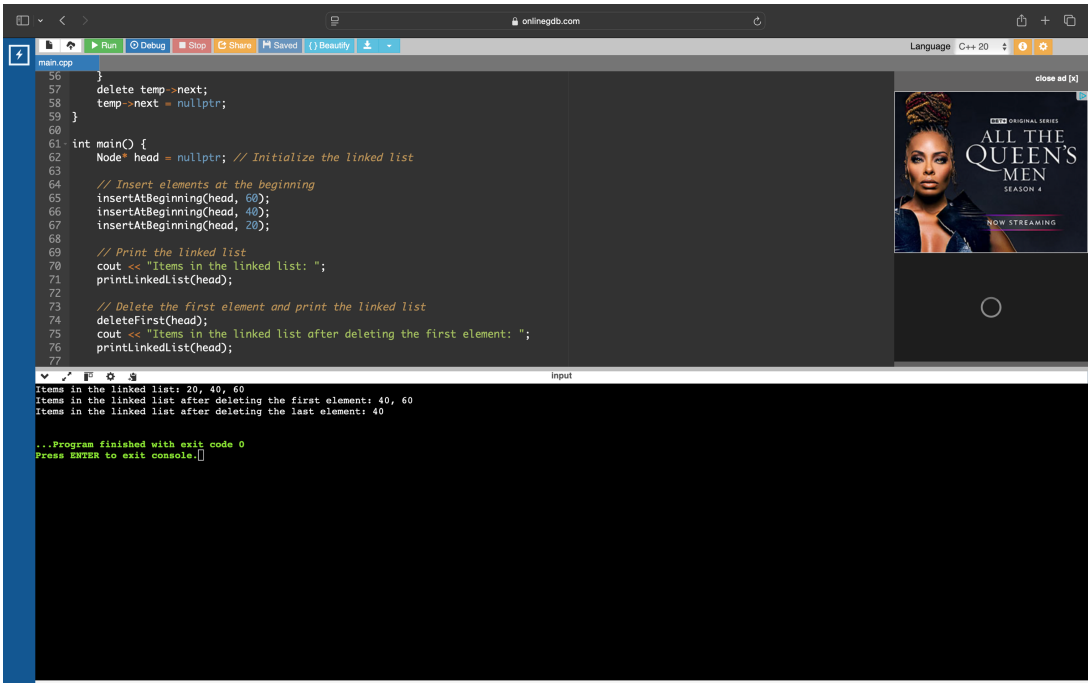
Lab Number - Section 09

CSCI 115 Lab 1 Report; Task 2

- Statement of Objectives
  - This lab experiment focuses on implementing and manipulating a linked list data structure. The primary objective is to understand the fundamental operations of a linked list, including insertion and deletion, and explore its practical applications through C++ programming. This report outlines the experimental procedure, analysis of results, challenges faced, and conclusions from this lab.

- Experimental Procedure:
  - To meet the objectives, the following procedure was followed:
    - Define the Linked List Structure: Implemented a singly linked list with a Node class containing data and a pointer to the next node.
    - Implement Operations: Developed functions to support insertion at the beginning, deletion of the first node, and deletion of the last node.
      - Insert at the Beginning: Added new nodes to the start of the list
      - Delete First Node: Removed the head node and updated the head pointer
      - Delete Last Node: Traversed the list to locate and remove the last node
    - Perform Tasks: Executed the following operations step-by-step:
      - Inserted nodes with values 60, 40, and 20 sequentially at the beginning
      - Printed the linked list contents
      - Deleted the first node and printed the updated list
      - Deleted the last node and printed the updated list
    - Validate Outputs: Ensured the results matched the expected output specified in the lab assignment

- Analysis:
  - The following results were obtained from the experiment:
    - Initial Linked List State: After inserting nodes, the linked list was printed as: 20 -> 40 -> 60
    - After Deleting the First Node: The head node was successfully removed and the linked list became: 40 -> 60
    - After Deleting the Last Node: The last node was successfully removed, and the linked list became: 40



- Encountered Problems:
  - The only thing I encountered is attempting to delete nodes from an empty list. I was resolved by adding checks to ensure the list was not empty before performing delete operations

- Conclusions
  - Through this lab, the following concepts were reinforced:
    - The implementation and manipulation of a linked list from scratch
    - The behavior of insertion and deletion operations, including handling edge cases
    - Debugging and error-handling techniques to manage null pointers and logical discrepancies.
      - This exercise provided a deeper understanding of the linked list data structure and its applications in computer science

- References
  - Lecture Slides on Stacks and Linked Lists